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Architecture of the microprocessors

Supervisor:

Prof. dr. ing. Mircea Giurgiu

Student:

Horgoș Raul-George

Group e_2031

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1. Project overview

A voice assistant is software that uses voice commands to perform routine tasks. It combines machine learning and artificial intelligence to recognize our voice and respond to our commands. Smartphones, tablets, smartwatches, smart speakers, and other gadgets all have voice assistant software. The most well-known are Bixby (Samsung), Siri (Apple), Alexa (Amazon), and Google Assistant (Google), which are taking over our homes because they work with air conditioners, washing machines, lightbulbs, and ovens, among other appliances. In this project i worked in a voice assistant that could be used in the automotive industry.

In this project i used an Raspberry Pi 400 and a cheap and small usb microphone good enough to realise the main ideea of the project: the voice assistant. The kit comes with the microSD card needed by the PCB, the mouse, a cable to power the whole kit and a HDMI cable to connect the board to a monitor so we can actually see what we are doing.

2. Problems encountered

One of the main problem encountered was the fact that everything was new to me, the raspberry kit and the project thematic. Despite anything to the contrary, the project seemed very hard at first, I managed to find solutions and make the project.

Other difficulties noticed were the libraries in pyhton used for the speech recognition, some of which were’t compatible with the version of the softwere on the Raspberry Pi board, or with the microphone.

3. Implementation

The final project work pretty straight forward, when we run the code, the console will execute the program and we will see the text “Ascult!” after which we can give it an instruction. If the microphone can’t hear anything it will give us an error, if the speech is not in Romanian or not intelligible we get the message “Nu s-a recunoscut!”. Otherwise the code runs great and we get a response for the input.

The code:

```
import speech_recognition as sr

r = sr.Recognizer()
my_mic = sr.Microphone()
#my_mic = sr.Microphone(device_index=1)
print("\nVoice assistant PORNIT!\n")

with my_mic as source:
    print("\nAscult!")
#    r.energy_threshold = 50
    r.adjust_for_ambient_noise(source)
    r.dynamic_energy_threshold = True
    audio = r.listen(source) #

#harvard = sr.AudioFile('harvard.wav')
#with harvard as source:
#    audio = r.record(source)

try:
    text = r.recognize_google(audio, language='ro-RO')
    print('\nAti spus "'+text+'"')
    if (text.lower() == "deschide geamurile"):
        print("\nS-au deschis geamurile!\n")
    elif (text.lower() == "închide geamurile"):
        print("\nS-au inchis geamurile!\n")
    elif (text.lower() == "pornește clima"):
        print("\nClima a fost pornita!\n")
    elif (text.lower() == "oprește clima"):
        print("\nClima a fost oprita!\n")
    elif (text.lower() == "aprinde lumina"):
        print("\nLumina a fost pornita!\n")
    elif (text.lower() == "închide lumina"):
        print("\nLumina a fost închisa\n")
    elif (text.lower() == "crește volumul radio"):
```

```
        print("\nVolumul radio a crescut!\n")
    elif (text.lower() == "scade volumul radio"):
        print("\nVolumul radio a scazut!\n")
    elif (text.lower() == "pornește radio"):
        print("\nRadio-ul a fost pornit\n")
    elif (text.lower() == "oprește radio"):
        print("\nRadio-ul a fost oprit ( ͡° ͜ʖ ͡°)")
except sr.UnknownValueError:
    print("\nNu s-a recunoscut\n")
```

4. Bibliography

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